SYDE 572 · INTRODUCTION TO PATTERN RECOGNITION · ASSIGNMENT 2

Linear Basis-Function Regression with Polynomial and Gaussian Bases

Assignment Report

Abstract—This report fits linear basis-function regression models to the Boston Housing data, predicting median house value (MEDV) from 13 town features. The first 100 observations are used for training and the remaining 406 for testing. Part 1 uses single-variable polynomial bases, examining overfitting, the degree-1 weights, one-dimensional fits, and L2 regularization selected by 10-fold cross-validation. Part 2 uses Gaussian bases centred on randomly chosen training points, examining the effect of the number of bases and of regularization.

Index Terms—linear regression, basis functions, overfitting, ridge regularization, cross-validation.

I. DATA AND SETUP

The Housing data contain 506 observations with 13 features and the target MEDV (in $1000s). Observations 1–100 form the training set and 101–506 the test set. Every feature is standardized as $z_j = (x_j - \mu_j)/\sigma_j$ using training-set statistics only, and the same transformation is applied to the test data. The feature CHAS is 0 for all 100 training rows, so its scale is set to 1 and its weight is unidentifiable (zero). Error is reported as root-mean-square error

$$\mathrm{RMS} = \sqrt{\frac{1}{N}\sum_i (y_i - \hat{y}_i)^2}.$$

Unregularized fits use least squares (`numpy.linalg.lstsq`); regularized fits solve $\mathbf{w} = (\Phi^\top\Phi + \lambda I')^{-1}\Phi^\top\mathbf{y}$, where I' leaves the bias unpenalized. All random choices use seed 572.

II. PART 1 — POLYNOMIAL BASIS FUNCTIONS

The basis is $(1, x_1, x_1^2, \ldots, x_1^M, x_2, x_2^2, \ldots, x_{13}^M)$: monomials of a single variable with no cross-terms, giving $1 + 13M$ weights for degree M.

A. Part 1(a): Training and Test Error versus Degree

TABLE I. PART 1(A): RMS ERROR VERSUS POLYNOMIAL DEGREE

Degree	Training RMS	Testing RMS
1	2.0372	36.44
2	1.5781	1097
3	1.4491	4.867e+04
4	1.2705	2.186e+07
5	1.0966	1.836e+10
6	1.0725	5.024e+12
7	**0.8815**	**6.879e+15**

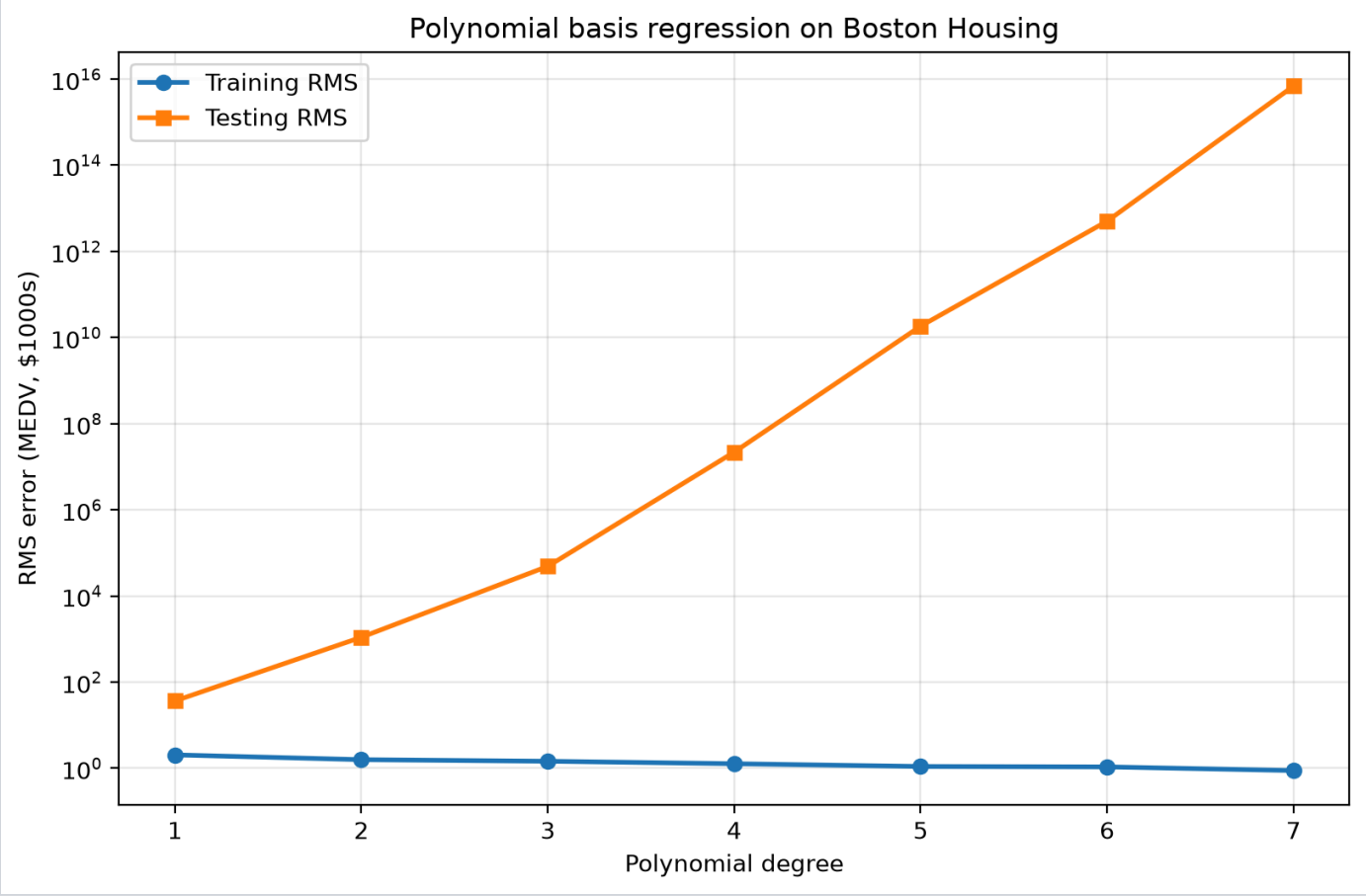


Fig. 1. Training and testing RMS for unregularized polynomial regression, degrees 1–7 (log vertical axis).

Training error falls steadily with degree, but test error grows by many orders of magnitude, a severe case of overfitting. Degree 7 has 92 weights for only 100 training points, so it fits the training sample closely. The first 100 rows are also not representative of the rest: after standardization with training statistics, some test features lie far outside the training range (CRIM reaches $|z| \approx 229$, RAD 14.5), and a degree-7 term on such a value is of order 10^{16}, which matches the observed test RMS. Even degree 1 has a test RMS of 36.4 for the same reason. Degree 1 is the best of those evaluated.

B. Part 1(b): Degree-1 Weights

TABLE II. PART 1(B): DEGREE-1 WEIGHTS (BIAS PLUS 13 STANDARDIZED FEATURES)

Index	Term	Weight	Train mean	Train std
0	bias	22.3090	—	—
1	CRIM	-1.3380	0.2833008	0.3871512704
2	ZN	0.1230	12.385	22.85167992
3	INDUS	-0.5311	6.6687	3.146500168
4	CHAS	0	0	0
5	NOX	1.0027	0.471321	0.04660157786
6	RM	4.1956	6.23441	0.4883777656
7	AGE	-2.1663	56.252	27.33905807
8	DIS	0.0107	5.177271	1.404997525
9	RAD	-0.1938	3.96	1.385063175
10	TAX	-0.3486	289.49	45.69190191
11	PTRATIO	0.2613	18.69	1.69041415
12	B	0.1811	385.3015	27.24965825
13	**LSTAT**	**0.1488**	**10.7729**	**5.671459653**

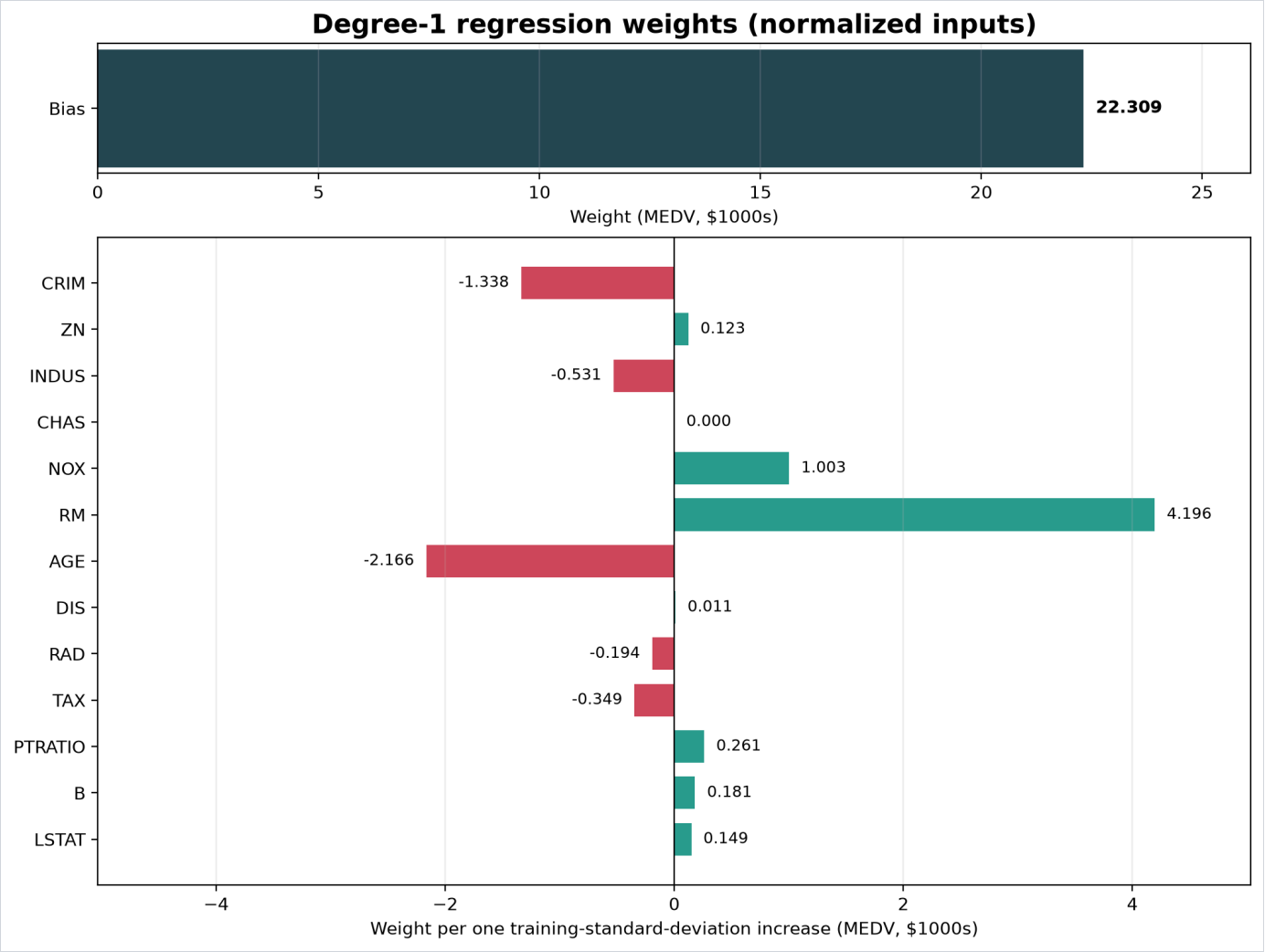


Fig. 2. Bias and standardized feature weights of the degree-1 model.

The weight on the 6th feature, RM (average rooms), is $w_5 = 4.196$. One training standard deviation of RM is 0.488 rooms, so this is about $4.196/0.488 \approx 8.6$ thousand dollars per extra room. The positive sign and large magnitude are reasonable, since larger dwellings are worth more. The weight on the 8th feature, DIS (weighted distance to employment centres), is 0.0107, effectively zero. This is plausible because DIS is correlated with other location variables and adds little once they are included. Coefficients are conditional and not causal; correlated predictors and the unrepresentative training split can produce unexpected signs. The bias (22.31) is the mean training MEDV.

C. Part 1(c): One-Dimensional Visualization (ZN)

Using only the standardized second feature ZN, polynomials of degree 1, 3 and 7 are fitted.

TABLE III. PART 1(C): ZN-ONLY POLYNOMIAL RMS

Degree	Training RMS	Testing RMS
1	5.695	9.352
3	5.665	9.434
7	**5.501**	**10.423**

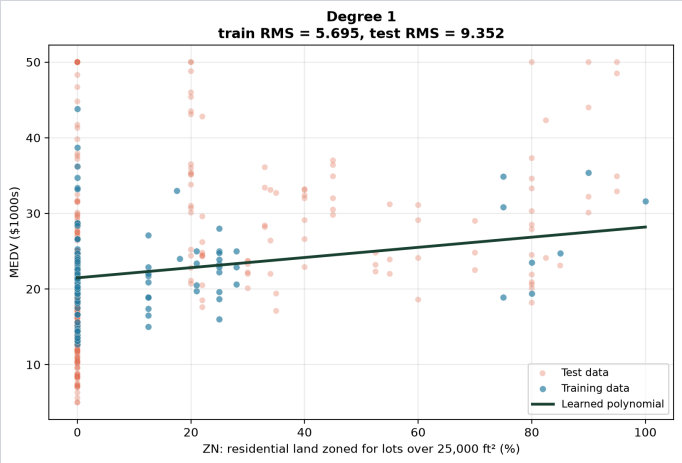


Fig. 3. Degree 1.

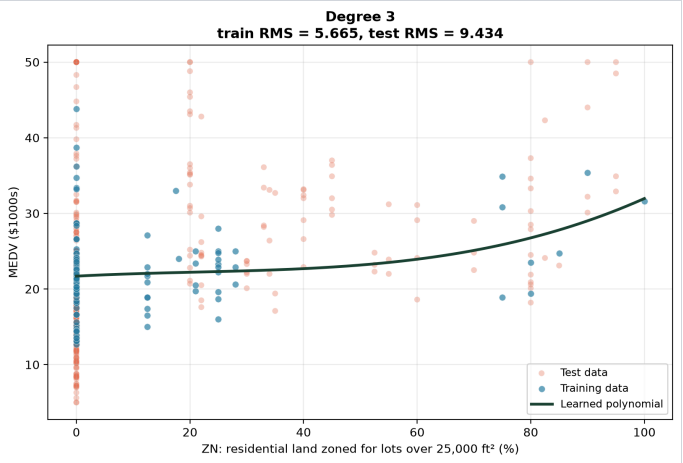


Fig. 4. Degree 3.

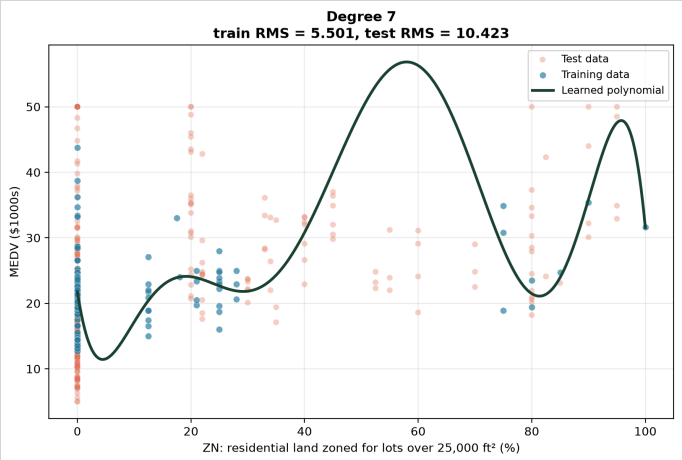


Fig. 5. Degree 7.

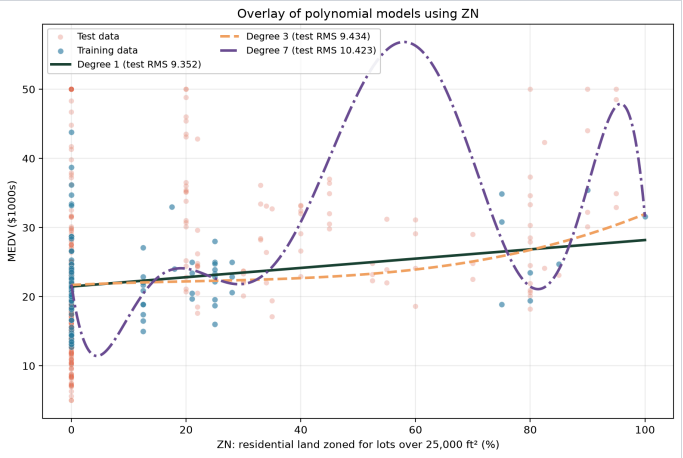


Fig. 6. Overlay of the three curves.

The degree-1 fit shows a modest positive trend, but points with the same ZN (especially ZN = 0) span a wide range of MEDV, so ZN alone explains little. Degree 3 adds mild curvature without improving test error (9.352 to 9.434). Degree 7 oscillates in sparsely sampled regions; its training RMS falls to 5.50 while test RMS rises to 10.42, again overfitting. Degree 1 is the most defensible.

D. Part 1(d): L2 Regularization with Cross-Validation

A degree-8 polynomial in ZN is fitted to the first 100 points, with $\lambda \in \{0, 0.01, 0.1, 1, 10, 100, 1000\}$ and 10-fold cross-validation. Normalization is re-estimated inside each fold from its 90 fitting points.

TABLE IV. PART 1(D): 10-FOLD CV, DEGREE-8 POLYNOMIAL ON ZN

λ	**Mean validation RMS**	**Std. error**
0	9.985	4.135
0.01	6.586	0.907
0.1	7.674	1.906
1	9.344	3.474
10	10.251	4.279
100	9.792	3.803
1000	**7.422**	**1.518**

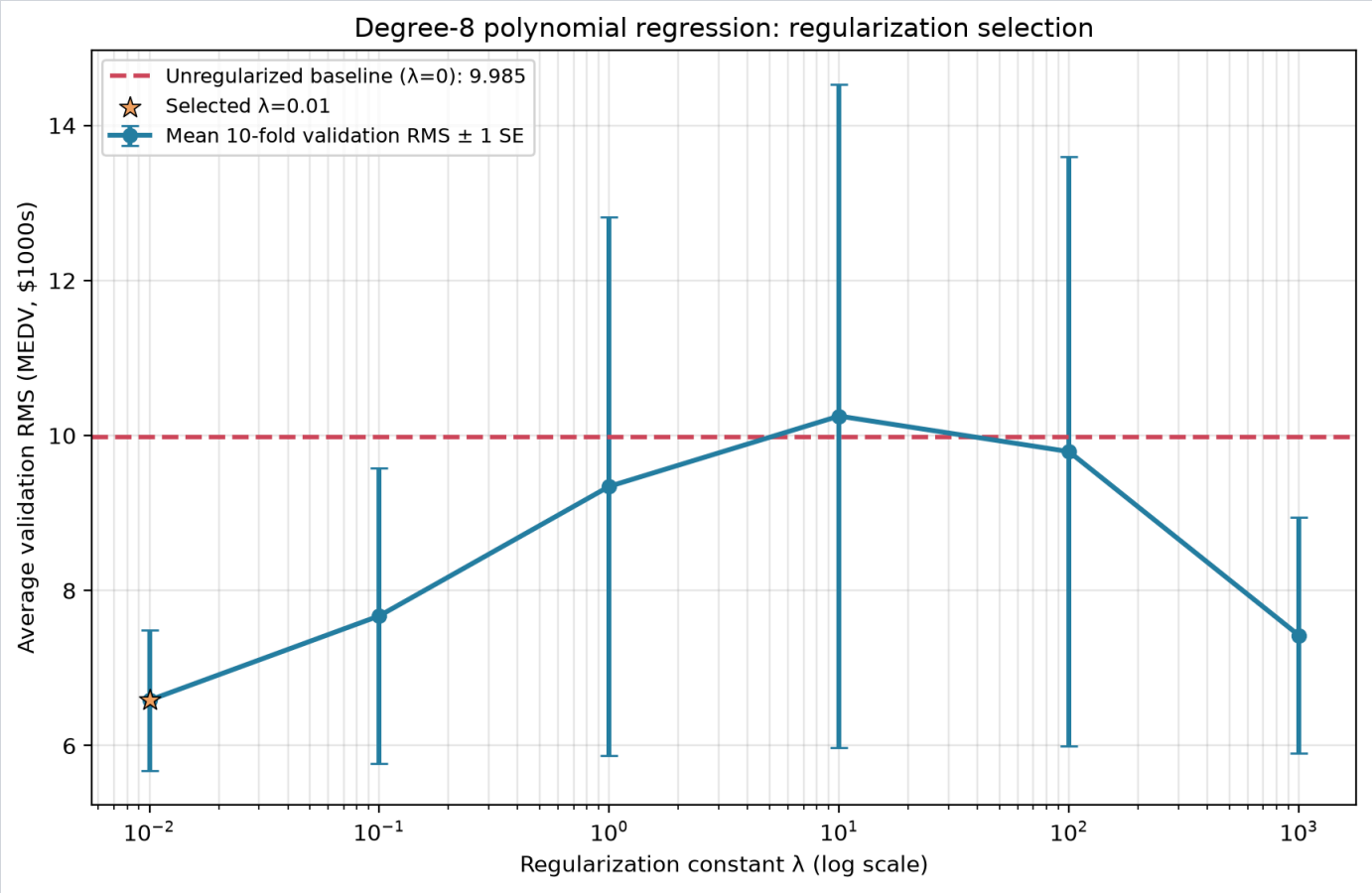


FIG. 7. Mean 10-fold validation RMS versus λ (log axis); error bars are one standard error; the dashed line is the $\lambda = 0$ baseline.

The lowest validation error (6.586) occurs at $\lambda = 0.01$, versus 9.985 unregularized, so I would choose $\lambda = 0.01$. Small shrinkage tames the high-variance degree-8 fit, while large penalties bias it toward a flat model. The curve is not monotonic and the error bars are wide because the dataset is small.

III. PART 2 — GAUSSIAN BASIS FUNCTIONS

The basis is $\phi_j(\mathbf{x}) = \exp(-\|\mathbf{z} - \boldsymbol{\mu}_j\|^2/2\sigma^2)$ with $\sigma = 1$ on all 13 standardized features, plus a bias. Centres $\boldsymbol{\mu}_j$ are the first K points of a random permutation of the training points.

A. Part 2(e): Number of Basis Functions

TABLE V. PART 2(E): GAUSSIAN BASIS, RMS VERSUS NUMBER OF BASES

K	Training RMS	Testing RMS
5	5.193	9.848
15	4.293	9.792
25	3.847	9.811
35	3.663	9.914
45	3.244	9.817
55	3.061	9.864
65	2.858	9.846
75	1.908	11.935
85	0.764	11.798
95	**0.299**	**11.755**

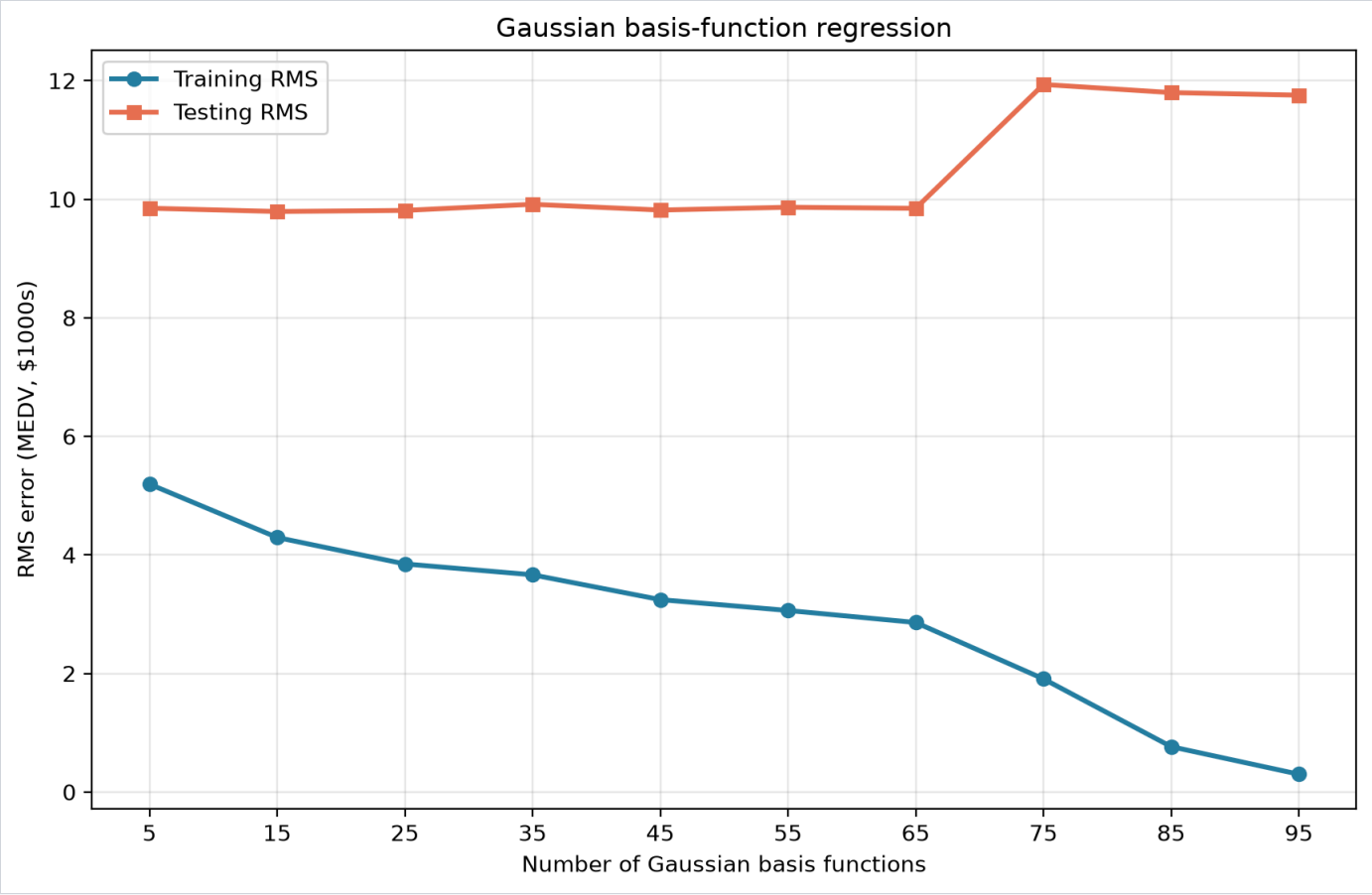


FIG. 8. Training and testing RMS for unregularized Gaussian regression, $K = 5, 15, \ldots, 95$.

Training error decreases monotonically from 5.19 to 0.30. Test error stays near 9.8 for $K \leq 65$ (minimum 9.79 at $K = 15$), then jumps to about 11.8–11.9 for $K \geq 75$, as the model (up to 96 weights for 100 points) nearly interpolates the training data. This is overfitting. Exact values depend on the random centres and on σ.

B. Part 2(f): L2 Regularization with Cross-Validation

With 90 basis functions and all features, 10-fold cross-validation is run per λ. In each fold, normalization and the 90 centres come from the fitting points only.

TABLE VI. PART 2(F): 10-FOLD CV, 90 GAUSSIAN BASES

λ	**Mean validation RMS**	**Std. error**
0	3.999	0.487
0.01	3.840	0.440
0.1	3.731	0.426
1	4.026	0.442
10	4.998	0.412
100	5.635	0.401
1000	**5.789**	**0.402**

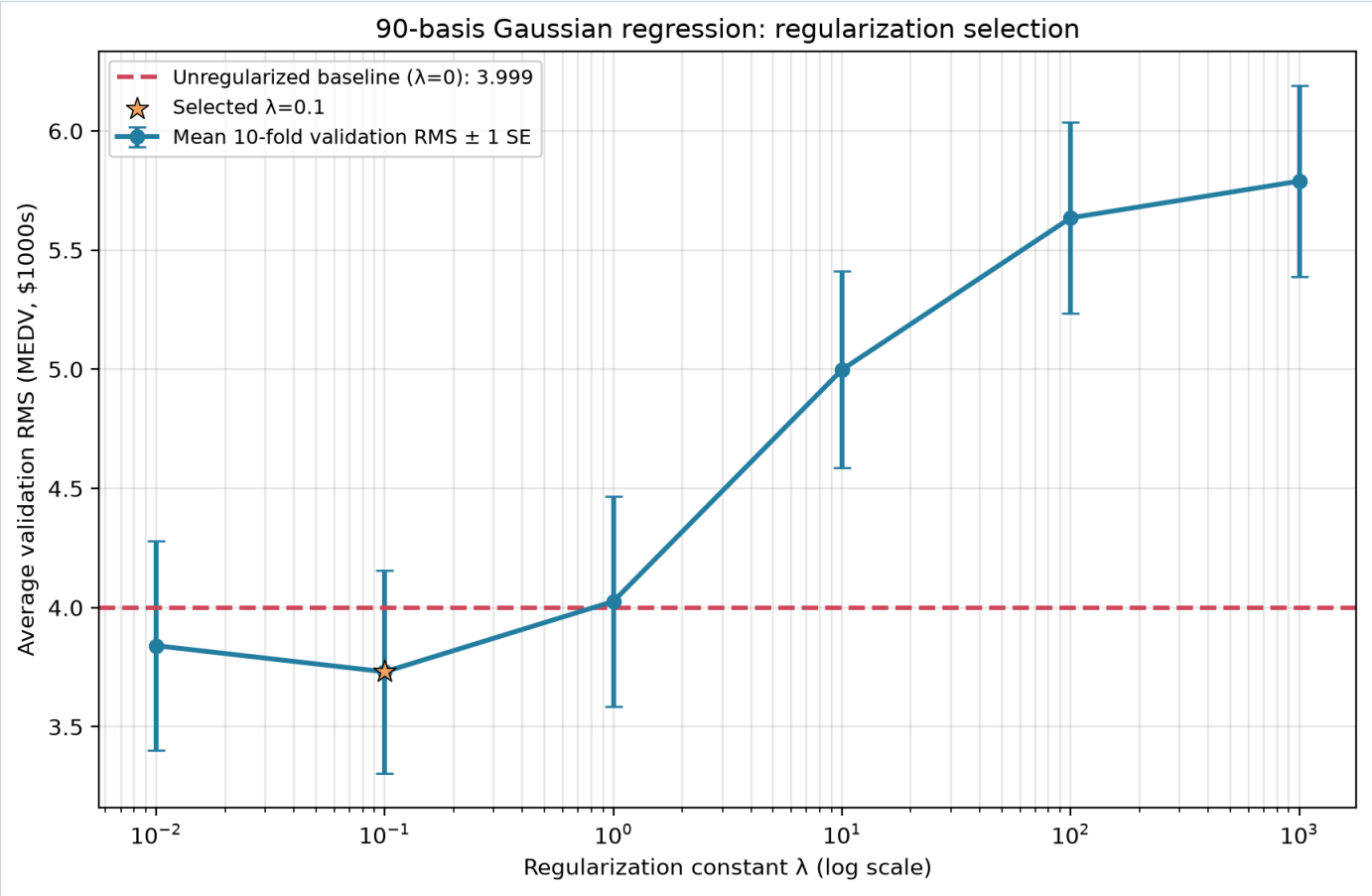


Fig. 9. Mean 10-fold validation RMS versus λ for 90 Gaussian bases (log axis, one-standard-error bars, dashed $\lambda = 0$ baseline).

The minimum validation RMS (3.731) is at $\lambda = 0.1$, so I would choose $\lambda = 0.1$. Modest shrinkage improves on the unregularized 3.999, while $\lambda \geq 10$ underfits.

IV. SUPPORTING FILES

Assignment handout | **part1a.py** | **part1b.py** | **part1c.py** | **part1d.py** | **part2e.py** | **part2f.py**

Run from this folder with `python part1a.py` etc.; each script writes its figure and CSV to `output/`.